

Three product and measure questions answered under $\diamond$

Run `fa_banach_001`, lane 9
`agent_lane_09`

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Status. Literature already answered: relative-consistency negative answers under Jensen's $\diamond$. This is not an absolute ZFC counterexample.

Source questions

Plebanek and Sobota, *Countable tightness in the spaces of regular probability measures*, arXiv:1405.2527 (2014), ask:

1. in Main Problem 1.1, whether countable tightness of $P(K)$ forces every $\mu \in P(K)$ to have countable Maharam type;
2. in Problem 5.1, whether countable tightness of $P(K)$ forces countable tightness of $P(K \times K)$;
3. in Problem 5.4, whether Corson's property (C) for $C(K)$ forces property (C) for $C(K \times K)$.

Supporting answer

Koszmider and Silber, *Countably tight dual ball with a nonseparable measure*, arXiv:2312.02750 (2023), prove in Theorem 1, assuming $\diamond$, that there is a compact Hausdorff space K such that $P(K)$ has countable tightness but K carries a Radon probability measure of uncountable Maharam type. This directly answers Question 1 negatively.

The first corollary immediately after their theorem states that the same K satisfies

$$P(K) \text{ has countable tightness,} \quad P(K \times K) \text{ has uncountable tightness.}$$

This is exactly a negative answer to source Problem 5.1.

Their next corollary states

$$C(K) \text{ has property (C),} \quad C(K \times K) \text{ does not have property (C),}$$

which is exactly a negative answer to source Problem 5.4.

The identification is explicit rather than retrospective: the supporting authors cite the Plebanek–Sobota result, explain that their example separates the tightness of $P(K \times K)$ from that of $P(K) \times P(K)$, and identify the property-(C) corollary as answering Pol's question.

Scope and remaining questions

The construction requires $\diamond$. The supporting paper also recalls Fremlin’s consistency result that such examples need not exist, so the packet records relative consistency and not a ZFC disproof.

Source Problem 5.3, asking whether property (C) of $C(K)$ implies countable tightness of $P(K)$, is explicitly said in the supporting paper to remain open in ZFC. The source question concerning Fréchet–Urysohn spaces of measures and countably determined measures is not settled by this construction.

Provenance and files

The local packet contains an exact-source local rendering of arXiv:1405.2527 as `source_paper.pdf` (the direct PDF fetch timed out twice) and the decisive 2023 paper as `supporting_paper_2312.02750.pdf`. The source’s Main Problem 1.1 duplicates an earlier formulation already indexed by this run; the two product questions are the additional exact answers recorded here.